

HINTS AND SOLUTIONS

- Q.1 B
Q.2 B
Q.3 B
Q.4 D
Q.5 B
Q.6 A
Q.7 B
Q.8 B
Q.9 B
Q.10 A
Q.11 B

Sol._{NB} $i_2 = i_3 = I_b$
 $V_1 = (V_2 + V_3) = V$
 and $P_2 = P_3$
 $\Rightarrow R_2 = R_3$
 and $V_2 = V_3 = V/2$
 $P_1 = P_2 = P_3 = P_4$ (given)
 $P_1 = \frac{V^2}{36}$; $P_3 = \frac{(V/2)^2}{R_3}$
 As $P_1 = P_3 \Rightarrow R_3 = 9\Omega$
 Also $R_3 = R_2 = 9\Omega$ **Ans.**

$$I_a = \frac{I}{3} \text{ and } I_4 = I$$

$$P_1 = P_4 = 4W$$

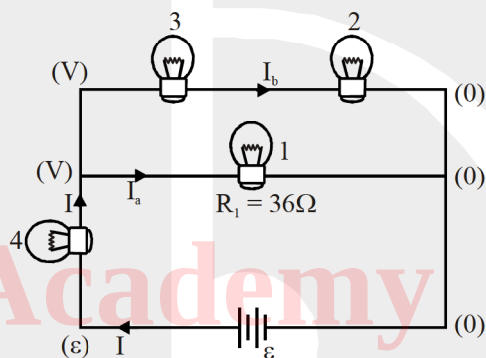
$$\left(\frac{I}{3}\right)^2 R_1 = (I)^2 R_4 = 4W$$

$$\Rightarrow \frac{R_1}{9} = R_4 \Rightarrow 4\Omega = R_4 \quad \text{Ans.}$$

$$\text{Also } I = 1A$$

$$R_{eq} = 16\Omega \text{ and } I = 1A$$

$$\varepsilon = 16V \quad \text{Ans.}$$



Q.12 C

Q.13 ABC

Sol. Potential drop across wire $V = \left(\frac{E_1}{r + R_1} \right) R_1$

R_1 : Resistance of wire

for the direction shown of current $E_2 > V$

so correct options are (A), (B) & (C)

Q.14 ACD

Sol. $i = \frac{24}{3+9+6} = \frac{4}{3} \text{ A}$

$$V_1 = \frac{4}{3} \times 9 = 12$$

$$V_2 = \frac{4}{3} \times 6 = 8 \text{ V}$$

at $t = \infty$

$$V_2 = V_1 = 24$$

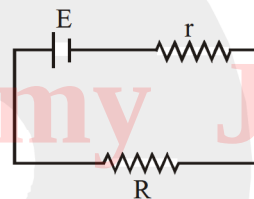
Q.15 ABD

Sol. At max power dissipation, $r = R$

so $i = \frac{E}{2R} = 5 \text{ A}$

and $i^2 R = 5 \text{ watt}$ solving $R = r = 0.2 \Omega$
 $E = 2 \text{ V}$

Also at $i = 2 \text{ A}$, $P = i^2 R$



Q.16 [Ans. (A) R, T (B) P (C) P, Q (D) R, T]

Sol. $R_1 = \frac{\rho l}{A} = R \text{ (say)}$ $R_2 = \frac{2\rho \cdot 2l}{\frac{A}{2}} = \frac{8\rho l}{A} = 8R$

Case 1: S_1 is open, S_2 is closed
all resistance are active.

$$R_{eq} = 5R$$

$$i_1 = \frac{\varepsilon}{5R} \text{ (current through } R_1) - \text{it is also the reading of ammeter}$$

$$i_2 = \frac{i_1}{2} = \frac{\varepsilon}{10R} \text{ (current through } R_2)$$

$$\text{Voltmeter reading} = i_2 R_2 = \frac{\varepsilon}{10R} \times 8R = \frac{4\varepsilon}{5}$$

$$\text{Current density in } R_1, J_1 = \frac{i_1}{A}$$

$$\text{Current density in } R_2, J_2 = \frac{\frac{i_1}{2}}{\frac{A}{2}} = \frac{i_1}{A} \Rightarrow J_1 = J_2$$

$$\text{Drift speed, } V_2 = \frac{J}{ne} \Rightarrow \text{drift speed in } R_1 \text{ is greater than drift speed in } R_2$$

$$\text{Electric field, } E = J \cdot \rho \Rightarrow E \text{ is greater in } R_2 \text{ than in } R_1.$$

$$\text{Power dissipated per unit volume} = \frac{i^2 R}{1A} = \frac{i^2 R}{A \cdot 1A} = \frac{i^2 \rho}{A^2} = \rho J^2$$

$$\Rightarrow \text{Power dissipated per unit volume is greater in } R_2 \text{ than in } R_1.$$

Case II: S_1 is closed and S_2 is open
Only resistance in circuit is $R_2 = 8R$

$$\text{Ammeter reading} = \frac{\varepsilon}{8R}$$

$$\text{Voltmeter reading} = \varepsilon$$

comparison is meaningless.

Case III: S_1 and S_2 both are closed
 R_1 is by-passed.

$$R_{eq} = \frac{R_2}{2} = 4R$$

$$\text{Ammeter reading} = \frac{\varepsilon}{4R}$$

$$\text{Voltmeter reading} = \varepsilon$$

comparison is meaningless.

Case IV: S_1 and S_2 both are open.
 $R_{eq} = R + 8R = 9R$

$$\text{Ammeter reading} = \frac{\varepsilon}{9R}$$

$$\text{Voltmeter reading} = \frac{8}{9} \varepsilon$$

$$\text{Current density in } R_1, J_1 = \frac{i}{A}$$

$$\text{Current density in } R_2, J_2 = \frac{i}{\frac{A}{2}} = 2J_1$$

$$\text{Drift speed} = \frac{J}{ne} \Rightarrow \text{Drift speed is same in } R_1 \text{ and } R_2$$

$$\text{Electric Field, } E = J\rho$$

$$\Rightarrow \text{electric field is greater in } R_2 \text{ than in } R_1$$

$$\text{Power dissipated per unit volume} = \rho J^2$$

$$\Rightarrow \text{It is greater in } R_2 \text{ than in } R_1. \text{ Ans.}]$$

Q.17 Ans. $\frac{2\pi a\lambda(8+\pi)}{32+\pi^2+16\pi}$

Q.18 Ans. $2\mu\text{C}$

Q.19 Ans. (i) E/R_1 (ii) $E/(R_1 + R_3)$

Q.20 [Ans. 0020]

Sol. All the elements of circuit are in parallel arrangement

$$\frac{1}{R_{\text{eq}}} = \frac{1}{40} \times \frac{1}{40} \times \frac{1}{40} \times \frac{1}{40} \times \frac{1}{20} \times \frac{1}{20}$$

$$= \frac{4}{40} \times \frac{2}{40}$$

$$R_{\text{eq}} = 5\Omega$$

$$\text{Power} = V^2/R = 20 \text{ W}$$

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